

HEALTH CARE

Health care courses offer students an opportunity to investigate factors contributing to personal health and, at the same time, gain an introduction to a range of careers in the health care industry. These careers include, but are not limited to, child care worker, dental assistant/dentist, gerontologist, laboratory technician/technologist, doctor, nursing assistant/nurse, pharmacy assistant/pharmacist, and personal support worker. Courses in health care help prepare students for rewarding careers in a sector that is set to expand rapidly as our population ages.

Students will gain hands-on experience using industry-standard instruments, equipment, and materials and practising current techniques. They will learn the professional terminology of the field, acquire transferable problem-solving skills, and expand their communication and interpersonal skills as they interact with their peers and clients in a variety of simulated care scenarios. As they acquire theoretical understanding and practical skills in their areas of interest, students will become better prepared to make informed career choices.

The list of approved emphasis areas for health care can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Health Care, Grade 11

University/College Preparation

TPJ3M

This course enables students to develop their understanding of basic health care procedures, including the safe use of appropriate instruments, equipment, and materials. Students will focus on health care fundamentals, including the anatomical features and physiology of the major body systems and the factors that affect homeostasis in the human body. Students will develop an awareness of health and safety issues in the health care field, analyse environmental and societal issues related to health care, and learn about professional practice standards and career opportunities in the field.

Prerequisite: None

A. HEALTH CARE FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of health care terminology and its correct usage;
- A2.** demonstrate a basic understanding of human anatomy and physiology;
- A3.** demonstrate an understanding of homeostasis and its relationship to personal health;
- A4.** describe the relationship between lifestyle choices and personal health and well-being;
- A5.** compare conventional and complementary methods of disease prevention and treatment.

SPECIFIC EXPECTATIONS

A1. Health Care Terminology

By the end of this course, students will:

- A1.1** demonstrate an understanding of health care terminology (*e.g., manifestation, obstruction*), and use it correctly in oral and written communication;
- A1.2** demonstrate an understanding of terminology related to human physiology (*e.g., hormone, cerebral vascular accident [stroke]*), and use it correctly in oral and written communication;
- A1.3** demonstrate an understanding of anatomical terminology (*e.g., proximal, hypochondrium*), and use it correctly in oral and written communication;
- A1.4** identify and use medical abbreviations correctly (*e.g., pt [patient], ECG [electrocardiogram]*);
- A1.5** name and identify the purpose of commonly used health care equipment (*e.g., X-ray machine, phlebotomy equipment, refraction equipment, walker frames, autoclave*);
- A1.6** pronounce and spell key health care terms correctly (*e.g., febrile = feb-ril; distal = dis-tal*).

A2. Anatomy and Physiology

By the end of this course, students will:

- A2.1** demonstrate an understanding of the basic structure of the human body (*e.g., cells, tissues, organs, systems*);
- A2.2** identify the four main types of tissue (*i.e., muscle tissue, nervous tissue, epithelial tissue, connective tissue*) and describe their basic functions;
- A2.3** demonstrate a basic understanding of the integumentary system (*e.g., skin layers and their components, sensory perception*);
- A2.4** demonstrate a basic understanding of the musculoskeletal system (*e.g., types of movement such as flexion, extension, rotation, abduction*);
- A2.5** demonstrate a basic understanding of the cardiovascular system (*e.g., by building a model of the heart to show major anatomical features and blood flow*);
- A2.6** identify the major structures of the lymphatic system and explain how it is related to the cardiovascular and immune systems (*e.g., lymphatic vessels and lymph nodes remove bacteria to protect the body*);
- A2.7** demonstrate a basic understanding of the respiratory system (*e.g., by building a functional respiratory model to illustrate the mechanism of breathing*);
- A2.8** demonstrate a basic understanding of the digestive system (*e.g., by building a model to illustrate the digestive pathway*);
- A2.9** identify the three main areas of the neurological system (*i.e., cerebrum, cerebellum, brain stem*) and describe their function;
- A2.10** demonstrate an understanding of visual indicators of health status that should be noted when carrying out a health assessment of an individual (*e.g., skin appearance [normal, bruised, jaundiced], breathing, posture, height, weight, alertness, balance, gait, signs of malnourishment*).

A3. Homeostasis

By the end of this course, students will:

- A3.1** describe the various balances that the body regulates in order to maintain a stable internal environment (*e.g., temperature, blood glucose, acid/base, blood pressure, fluid balance*);
- A3.2** identify factors that can create a homeostatic imbalance (*e.g., vomiting/diarrhea can lead to dehydration and electrolyte imbalance*);
- A3.3** explain how imbalances in body systems affect personal health (*e.g., an imbalance in blood glucose can lead to hypoglycemia/hyperglycemia*).

A4. Lifestyle Choices

By the end of this course, students will:

- A4.1** identify and describe lifestyle choices that can improve a person's health and well-being (*e.g., avoidance of substance use/abuse, stress management, regular exercise, prevention of dental cavities, safe use of MP3 players and cell phones, using an Aboriginal medicine wheel as a health framework*);
- A4.2** assess a simulated client's dietary strengths and weaknesses and make appropriate food and nutrition suggestions, using appropriate resources (*e.g., Eating Well with Canada's Food Guide [2007]; Eating Well with Canada's Food Guide: First Nations, Inuit and Métis; cultural adaptations of Canada's Food Guide available from the Ontario Public Health Association*);

- A4.3** identify and describe different types of community health and social services that help people maintain active and healthy lives (*e.g., community care access centres, nursing, mental health services, physiotherapy, lab services, Meals on Wheels, vision services, dental services, pharmacy services*).

A5. Disease Prevention and Treatment

By the end of this course, students will:

- A5.1** research and describe conventional medical treatments for disease, disability, or injury (*e.g., cancer – radiation treatment, chemotherapy, surgery; cholelithiasis – diet, lithotripsy, microscopic surgery, pain control*);
- A5.2** research and describe complementary medical treatments for disease, disability, or injury (*e.g., acupuncture for pain management, First Nations ritual of smudging to purify and cleanse the body, therapeutic touch*);
- A5.3** describe a variety of conventional and complementary methods of preventing and treating disease that can be combined to provide a holistic approach (*e.g., prescribed and over-the-counter medication, meditation, herbal remedies, chiropractic treatment, naturopathy, massage therapy, surgery*);
- A5.4** demonstrate an understanding of the chain of infection and the principles of infection control.

B. HEALTH CARE SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use health care instruments, equipment, and materials safely and correctly;
- B2.** demonstrate the ability to use vital signs to determine a client's health status;
- B3.** demonstrate the ability to apply health care skills and techniques safely and to industry standards;
- B4.** demonstrate the ability to apply a variety of techniques for communicating with clients and collecting client information.

SPECIFIC EXPECTATIONS

B1. Instruments, Equipment, and Materials

By the end of this course, students will:

- B1.1** identify common medical instruments (*e.g., sphygmomanometer, forceps, double-ended explorer*), equipment (*e.g., Hoyer lift, intravenous [IV] pole, weigh scales, microscope, mouth mirror*), and materials (*e.g., wound dressings, bed linens and lift sheets, gloves, protective gowns, eye shields*), and use them safely and correctly;
- B1.2** disinfect instruments and equipment, using accepted medical aseptic procedures, to prevent the spread of infection (*e.g., nosocomial infection secondary to contaminated medical equipment/devices*);
- B1.3** prepare, clean, and store instruments and equipment correctly, following standard industry procedures (*e.g., calibrate sphygmomanometer correctly prior to use, prepare instruments for autoclaving, clean pill cutter before and after use*).

B2. Vital Signs

By the end of this course, students will:

- B2.1** demonstrate the ability to measure the four primary vital signs (temperature, pulse, respiration, blood pressure) and to assess degree of pain (often considered an additional vital sign);
- B2.2** assess and document a client's vital signs: temperature (by oral, axillary, and tympanic routes); pulse rate (rhythm and volume by apical and radial sites); respiration rate (rhythm

and character); blood pressure; and pain intensity (*e.g., using a pain scale*);

- B2.3** identify normal values for temperature, pulse, respiration, and blood pressure across the lifespan, and assess possible health implications of abnormal values (*e.g., elevated temperature suggests a possible infection; an elevated blood pressure reading may be an early indication of hypertension*).

B3. Skills and Techniques

By the end of this course, students will:

- B3.1** demonstrate an understanding of correct hand hygiene procedures (*e.g., knowing when they are required, knowing when to use hand washing rather than antibacterial sanitizers*), and apply as required;
- B3.2** perform common caregiving skills and techniques (*e.g., bed making, bed bathing, feeding, toileting, oral care, weighing*) safely and correctly, using baby and/or adult mannequins;
- B3.3** perform mobility techniques (*e.g., turning, lifting, and transferring clients; using crutches or walkers and instructing clients in their use*) safely and correctly;
- B3.4** perform a range of motion and positioning techniques safely and correctly to prevent the development of decubitus ulcers and contractures in clients;
- B3.5** demonstrate an understanding of (*e.g., through role play*) and apply health and safety procedures relating to proper body mechanics and ergonomics (*e.g., base of support, good body posture*);

B3.6 perform basic physical assessment skills
(e.g., assessment of visual acuity using a Snellen Chart; assessment of hearing, breath sounds, bowel sounds, gait and balance; neurological assessment).

B4. Communication and Documentation

By the end of this course, students will:

B4.1 demonstrate the ability to use therapeutic communication techniques (e.g., active listening and paraphrasing) in a variety of client situations (e.g., clients recovering from a stroke, clients with dementia, clients dealing with pain and stressful situations);

B4.2 identify various communication barriers (e.g., barriers relating to health conditions, age, stage of development, and differing cultural communication styles) and apply strategies for overcoming them;

B4.3 demonstrate competence in the use of methods for collecting client data and documenting client information (e.g., role play how to interview a client to obtain a client history).

C. HEALTH CARE, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe the impact of health care industry activities on the environment and identify ways of minimizing their harmful consequences;
- C2.** describe the impact of current social patterns and trends on personal health and the delivery of health care.

SPECIFIC EXPECTATIONS

C1. Health Care and the Environment

By the end of this course, students will:

- C1.1** describe the potential impact on the environment of biohazardous waste from health care facilities (*e.g., body fluid and human tissue, sharps containing bacteria or viruses*);
- C1.2** identify safe methods for the handling, storage, and disposal of waste and biohazardous materials (*e.g., use of checklists, sharps containers, double wrapping, proper labelling*);
- C1.3** describe good environmental practices that can be applied in the health care industry (*e.g., using energy-efficient lighting; reducing, reusing, or recycling packaging material; storing information electronically instead of on paper*).

C2. Health Care and Society

By the end of this course, students will:

- C2.1** identify demographic and lifestyle patterns and trends (*e.g., an aging population, lack of exercise*) and describe their implications for the health care industry (*e.g., increased obesity and susceptibility to diabetes and cardiovascular disease, changes in care needs, changes in treatments and interventions, trends in mental illness and public attitudes towards it*);
- C2.2** research and analyse the relationship between socio-economic status and health (*e.g., differences in various health parameters among children aged 4 to 6 from different socio-economic backgrounds*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and comply with safe working practices and the laws and regulations governing the health and safety of workers in the health care industry;
- D2.** demonstrate an understanding of legal and ethical standards governing the practice of health care;
- D3.** describe career opportunities in the health care field and related educational requirements.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and comply with legislation, regulations, standards, and requirements pertaining to worker safety in the health care workplace (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** demonstrate an understanding of and apply procedures to ensure safe and productive work practices in the health care workplace (e.g., *use tools and equipment safely as outlined in safety manuals, operating instructions, and institutional requirements; conduct a safety inspection or audit of a facility; use a checklist to keep track of tools and equipment*);
- D1.3** use protective clothing and equipment as required to keep themselves and others safe and free from harm.

D2. Professional Standards

By the end of this course, students will:

- D2.1** identify and explain the principal components of legislation pertaining to health care in Ontario (e.g., *Regulated Health Professions Act [RHPA]* sets the framework for regulating the scope of practice of health professions);
- D2.2** explain the need for the professional and ethical standards that health care workers are expected to adhere to (e.g., *accountability, knowledge, competence, respect for patients' rights and privacy, respect for cultural and religious diversity, respect for the sanctity of life*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** identify various career opportunities in health care and describe the education and/or training required for entry into these occupations;
- D3.2** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the health care industry (e.g., *mentoring programs, virtual networking/support groups, specialized post-secondary programs, relevant industry associations*);
- D3.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the health care industry, as identified in the Ontario Skills Passport (e.g., *document use, decision making, finding information*);
- D3.4** demonstrate an understanding of and apply the work habits that are important for success in the health care industry, as identified in the Ontario Skills Passport (e.g., *working safely, teamwork, reliability*);
- D3.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in health care (e.g., *Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.